

B2 BACnet / Modbus

BACnet

March 4th – 5th, 2026

BACnet



MP  **BUS**
MP-BUS COMPATIBLE



 **Modbus**

TCP



RTU



 **BACnet**™

MSTP

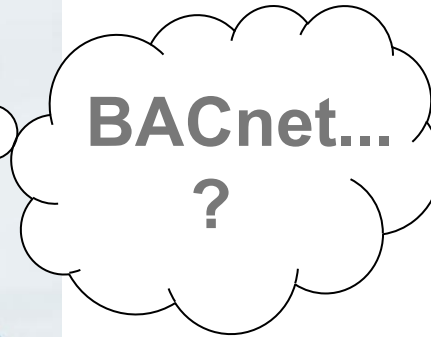
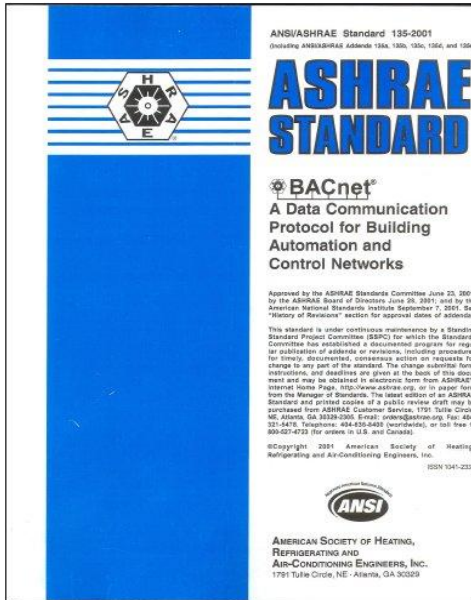


IP



KNX®





data communication protocol for Building Automation and Control networks

	standard	standard	Vendor ID	Vendor ID
Start	ANSI/ASHRAE 135	ISO 16484-5	423 BELIMO	1000 issued
1987	1995	2003	2010	2017

ASHRAE - official BACnet Web Site: www.bacnet.org

BACnet Interest Group (BIG) - in Europe: www.big-eu.org

BACnet International (BI) - International: www.bacnetinternational.org

How many Vendor IDs per Country:

as of 26.Juni 23: 1418 Vendor IDs issued

Vendor IDs – BACnet Committee

- USA: 472
- Germany: 159
- Canada: 92
- China: 77
- Japan: 69
- United Kingdom: 67
- Australia: 46
- Switzerland: 42
- Italy: 32
- France: 29
- Poland: 29
- India: 28
- Korea: 27
- Austria: 24
- Sweden: 21
- Netherlands: 19
- Denmark: 18
- Czech Republic: 16
- Taiwan: 13
- Belgium: 10
- Finland: 10



standard
ISO 16484-5

Vendor ID
423 BELIMO

Vendor ID
1000 issued

2003

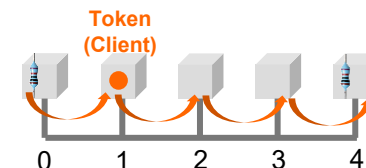
2010

2017

BACnet :Building Automation and Control networks

Fields	Heating, Ventilation, Air Conditioning, Lighting, Fire alarm, Intrusion and Access Control Systems
Application Building	BACnet is a protocol for buildings, originally used in Building Automation. BACnet MSTP is used at the Automation and Field Level .
System Setup	Client-Server MS/TP :Multiple Manager (MSTP: Master-Slave -Token-Passing – Manager - Subordinate -Token-Passing)
Actuator / Sensor	BACnet Server MS/TP: MSTP Manager
Medium	MS/TP: RS485/EIA485 (Max. length per Line without Repeater 1200m) IP: IP based
Topology	MS/TP: Bus Line with Terminating Resistor IP: IP Network
Max. Number of participants	MS/TP: Typically 32 Devices per Line Device-specific limits and possibilities check BACnet Interface Description!
Speed	MS/TP: 9'600 - 115'000 Baud, often 38'400 Bd, Network with low performance
Norms, Standards Links	Standardized device profiles of ASHRAE or AMEV Test and Certifications (BTL) https://www.bacnetinternational.org , https://www.bacnetlabs.org
Strengths	• Manufacturer-independent standard protocol with a wide range of functions • Standardization of the protocol • independent testing and certification bodies • All information about BACnet is simple to query • COV (Change Of Value): Automatic transmission of changed values possible
Weaknesses	• BACnet protocol requires more communication bandwidth in comparison to Modbus • BACnet Knowhow for Integration necessary

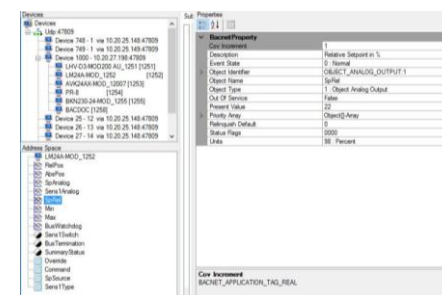
Bus Line:



BACnet Interface Description:



Tool YABE:



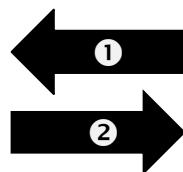
BACnet Objects and Services

BACnet Server

Actuator / Sensor

BACnet Objects:

Rel. Setpoint % (AI1)
Rel. Position % (AI2)
Setpoint Source (MV220)

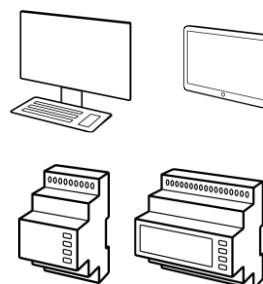


BACnet Services:

Read Data
Write Data
Value Change
Notification (COV)

BACnet Client

Automation Station / Management Station



DDC

The BACnet server (e.g. actuator) has the data which can be queried by the BACnet client (e.g. automation station) via services.

The data is organized in objects and properties.

BACnet Objects and Properties – Example: NM24A-MOD

- NM24A-MOD_7005
- ANALOG_INPUT:1
- ANALOG_INPUT:2
- ANALOG_INPUT:6
- ANALOG_INPUT:20
- ANALOG_OUTPUT:1
- ANALOG_VALUE:97
- ANALOG_VALUE:98
- ANALOG_VALUE:130
- BINARY_INPUT:20
- BINARY_INPUT:99
- BINARY_INPUT:101
- MULTI_STATE_INPUT:100
- MULTI_STATE_INPUT:106
- MULTI_STATE_INPUT:110
- MULTI_STATE_OUTPUT:1
- MULTI_STATE_VALUE:120
- MULTI_STATE_VALUE:122
- MULTI_STATE_VALUE:220

Object Type
⇒
Read Property:
Object Name

- NM24A-MOD_7003
- RelPos
- AbsPos
- SpAnalog
- Sens1Analog
- SpRel
- Min
- Max
- BusWatchdog
- Sens1Switch
- BusTermination
- SummaryStatus
- InternalActivity
- StatusActuator
- StatusDevice
- Override
- Command
- SpSource
- Sens1Type

RelPos (Analog Input Object)

BacnetProperty	
Cov Increment	1
Description	Relative Position in %
Event State	0 : Normal
Object Identifier	OBJECT_ANALOG_INPUT:1
Object Name	RelPos
Object Type	0 : Object Analog Input
Out Of Service	False
Present Value	50.73
Status Flags	0000
Units	98 : Percent

BACnet Services - BIBBs

Standard and used from Belimo



- Alarm & Event Services
- Object Access Services
- Network Security Services
- Remote Device Management Services
- File Access Services
- Virtual Terminal Services

Extract BIBBs from

BELIMO PICS

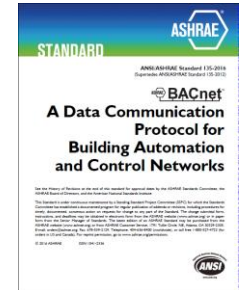
Data Sharing - ReadProperty-B (DS-RP-B)
Data Sharing - ReadPropertyMultiple-B (DS-RPM-B)
Data Sharing - WriteProperty-B (DS-WP-B)
Data Sharing - WritePropertyMultiple-B (DS-WPM-B)
Data Sharing - COV-B (DS-COV-B)
Device Management - DynamicDeviceBinding-B (DM-DDB-B)
Device Management - DynamicObjectBinding-B (DM-DOB-B)
Device Management - DeviceCommunicationControl-B (DM-DCC-B)

BACnet Interoperability Building Blocks



Extract BIBBs
from BELIMO PICS

Data Sharing - ReadProperty-B (DS-RP-B)
Data Sharing - ReadPropertyMultiple-B (DS-RPM-B)
Data Sharing - WriteProperty-B (DS-WP-B)
Data Sharing - WritePropertyMultiple-B (DS-WPM-B)
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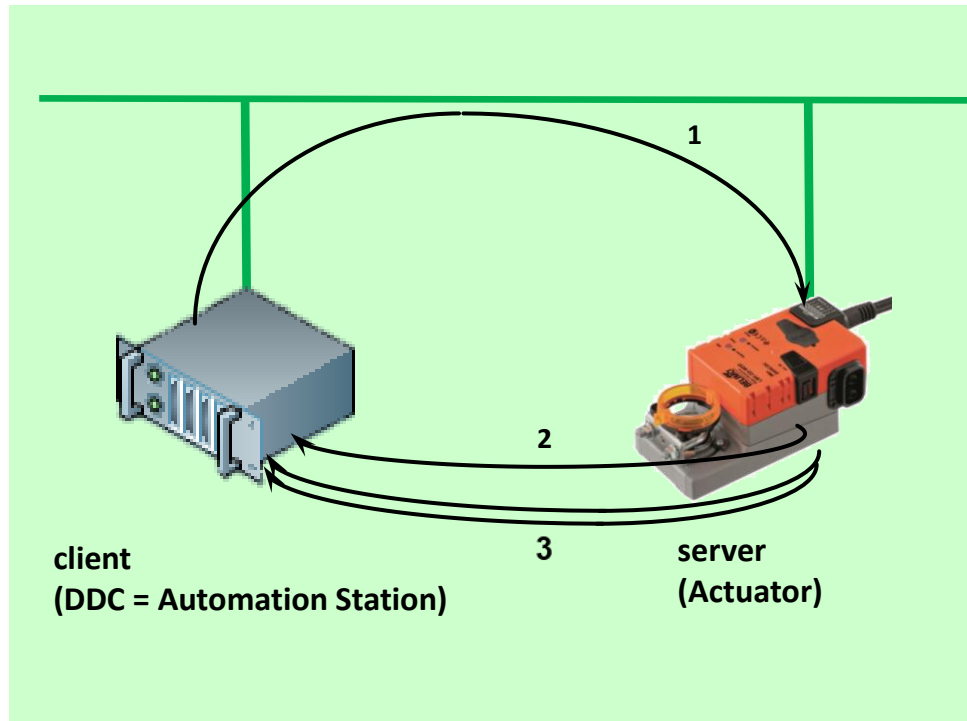
- ⇒ A: Client Function (Example: DDC)
- ⇒ B: Server Function (Example: Belimo Actuator)

- BIBBs Example **DS-RP-A** : Client Function
DS-RP-A (Data Sharing-ReadProperty-A)

BACnet Service	Initiate	Execute
ReadProperty	X	

BACnet COV - DS-COV-B

Change of Value (COV) Service



1. Client (DDC) subscribes to a value (datapoint) of the server (actuator)
2. Server (actuator) acknowledges this and sends the current value to the client (DDC).
3. After changing the value, the server (actuator) automatically sends the new value to the client (DDC).

Remark: Better than polling as network traffic is less.

Special Case with UK24BAC:

DS-COVU-B: with this Building Block, B is using unsubscribed COV Notifications, unsolicited

BACnet Interoperability Building Blocks



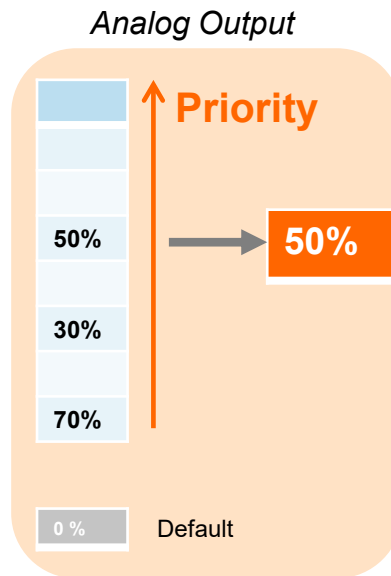
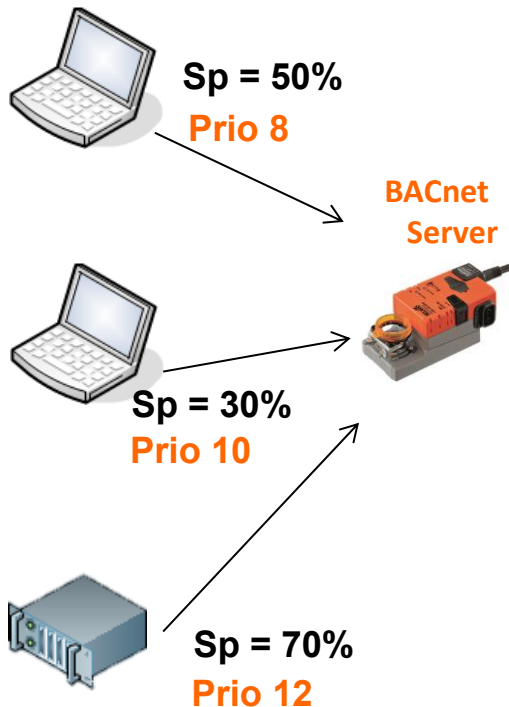
- ⇒ A: Client Function (Example: DDC)
- ⇒ B: Server Function (Example: Belimo Actuator)
- BIBBs Example **DS-COV-B** : Server Function
DS-COV-B (Data Sharing-COV-B)
 - (A) subscribes for value changes of a specific Object (B)
 - (B) sends (A) value changes

BACnet Service	Initiate	Execute
SubscribeCOV		X
ConfirmedCOVNotification	X	
UnconfirmedCOVNotification	X	

Commanding and Priority Array

3 x write Setpoint: Who wins?

BACnet Clients



Priority Level	Application
1	Manual-Life Safety
2	Automatic-Life Safety
3	Available
4	Available
5	Critical Equipment Control
6	Minimum On/Off
7	Available
8	Manual Operator
9	Available
10	Available
11	Available
12	Available
13	Available
14	Available
15	Available
16	Available
-	Relinquish Default

Standard BACnet Device Profiles

- B-OWS BACnet Operator Workstation, GLT
- B-AWS BACnet Advanced Operator Workstation (NEU)
- B-OD BACnet Operator Display (NEU)
- B-BC BACnet Building Controller
- B-AAC BACnet Advanced Application Controller
- B-ASC BACnet Application Specific Controller
- B-SA BACnet Smart Actuator
- B-SS BACnet Smart Sensor
-

Example Relinquish Default

VAV Compact: Bus Watchdog

from PICs

AV[130]	Timeout for Bus Watchdog in s 0 = watchdog deactivated. If the Present_Value is not ZERO, the implementation tracks write procedures to Present_Value of AO[1] and MO[1]. If the Present_Value of AO[1] or MO[1] is written, the timer is reset. Upon timeout the Priority_Array of the AO[1] is cleared and the Relinquish_Default becomes valid.	0...3600
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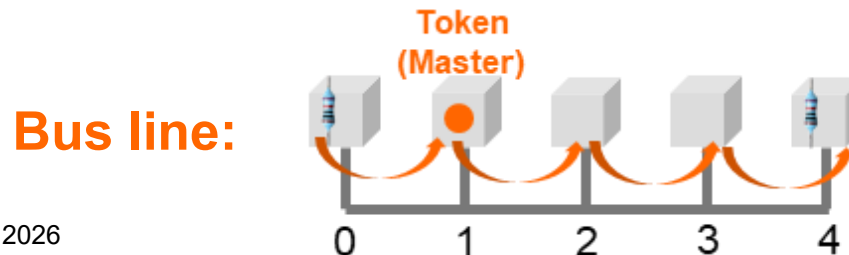
AO[1]	Relative Setpoint in %	0...100
-------	------------------------	---------

Relinquish Default = Default / start value of the Present Value

For future products different approach

MS/TP: RS-485 wiring

- Industry standard **EIA-485** for asynchronous data transmission
- Bus / line technology (Daisy Chain):
 - Stub lines generally not allowed
 - ⇒ **Therefore there are only short Actuator cable lengths for RS-485 communication such as Modbus RTU and BACnet MS/TP.**
- Longer cables must be terminated with a **terminating resistor** on both sides!
 - install terminating resistor physically
 - if supported Switch on terminating resistor by software (e.g. with ZTH / App)
- **MS/TP:**
 - Manager-Subordinate - Token Passing
 - We **only** use **managers**, all managers use the token for communication.



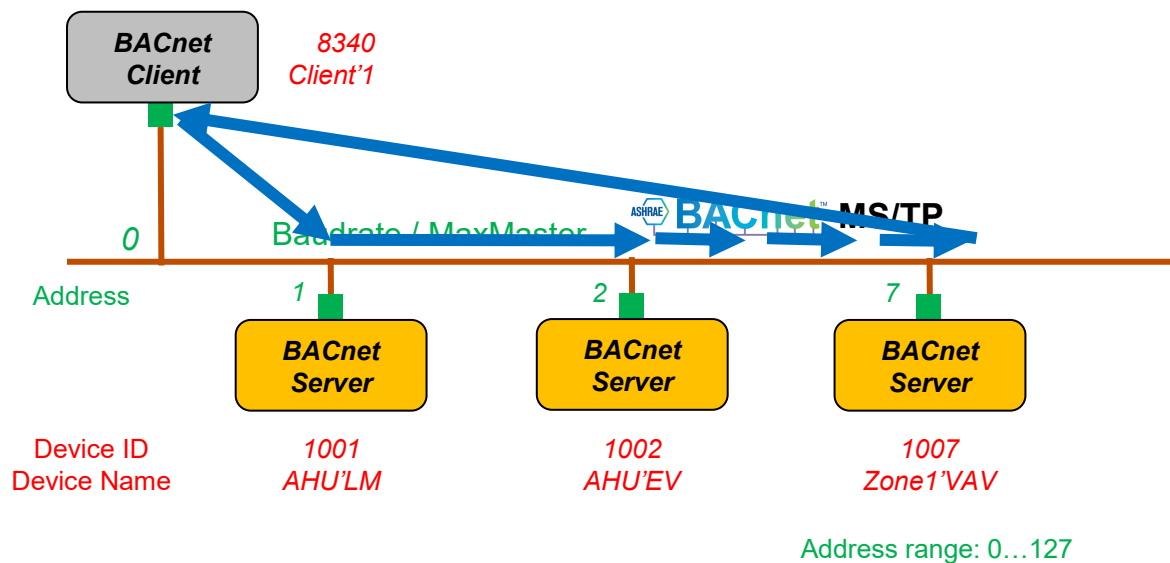
BACnet EDE File



How can BACnet objects of a device or system be exported to a file?

Engineering Data Exchange Template for BACnet Systems

- Excel file (CSV)
- Export from manufacturer specific engineering tool or browse and export with tool like yabe



BACnet MS/TP Addressing



BACnet Client - Yabe

InstNr : 100

MAC Addr = 0



InstNr : 1000

MAC Addr = 1



BACnet Router

Baud rate = 38'400
MaxMaster = 6

BACnet Client - Yabe

InstNr: 106

MAC Addr = 6



InstNr: 102

MAC Addr = 2



LM24A-MOD

InstNr: 103

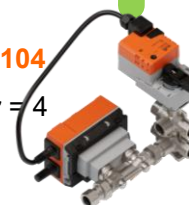
MAC Addr = 3



PRCA-BAC-S2-T

InstNr: 104

MAC Addr = 4



EP..R6+BAC

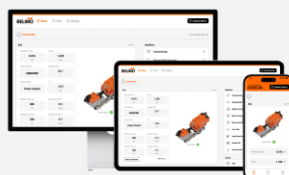
InstNr : 105

MAC Addr = 5



BKN230-24-MOD

Parameterisation



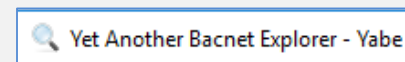
Belimo Assistant 2



LINK.10



ZTH EU



BACnet MS/TP Configuration for Belimo Devices

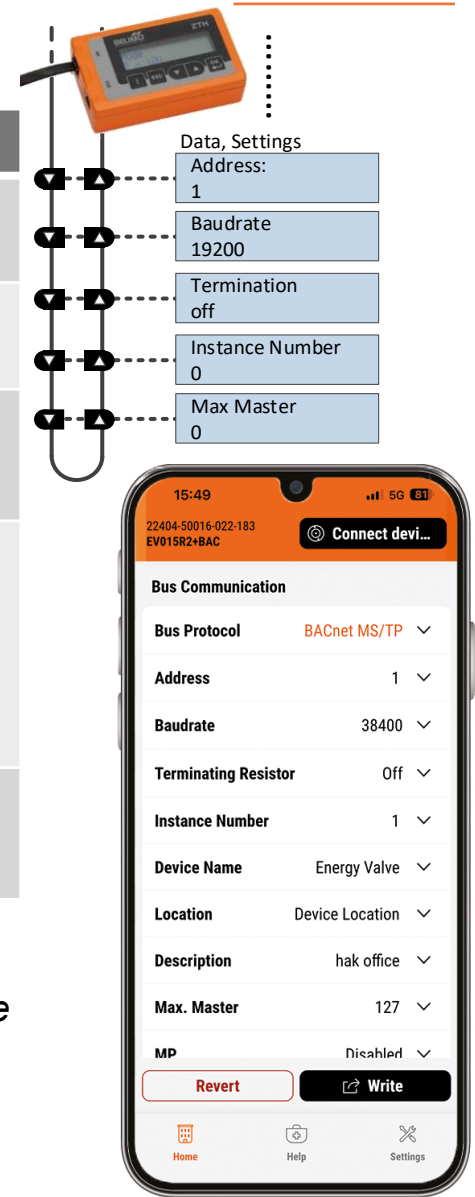
Device Name	Site4'Floor2 'Zone3'VAV1	BACnet Internetwork-wide unique Text for the Device Identification
Instance Number	1 - 4194302	BACnet Internetwork-wide unique Number for the Device Identification
MAC Address	0-127	MS/TP MAC Address , unique in each MS/TP Line
Baud rate	9'600 19'200 38'400 76'800 115'200	Baud rate of the RS485 Network . Insert the Terminating Resistor at the first and last device of the line. Supported baud rates of the respective device can be found in the PICS.
Max Master	0-127	Last MS/TP device address of the line. Optimizes the communication. Set to highest MAC address of the MS/TP line.



Check max. number of devices per MS/TP Line (typically 32 devices)



BACnet Instance Number and Device Name must be unique in the entire BACnet Internetwork!



LM24A BACnet Interface Description



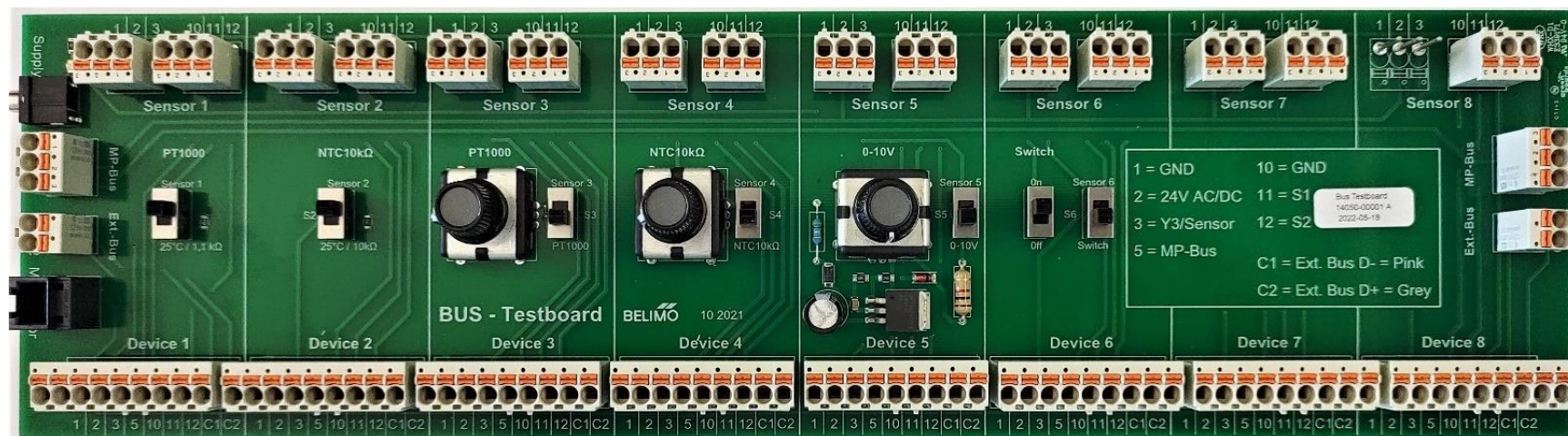
DDC Integration

DDC Integration



- Placeholder -> PPT

Workshop I

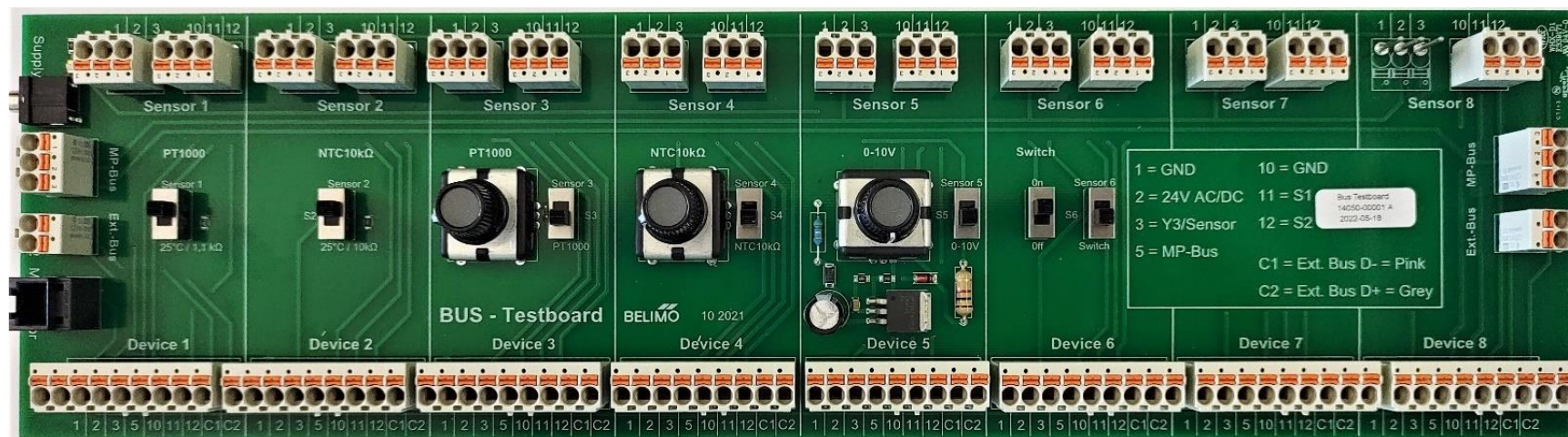


Workshop I



- Placeholder -> PPT

Workshop II



Workshop II



- Placeholder -> PPT

BACnet IP



MP  **BUS**
MP-BUS COMPATIBLE



 **Modbus**

TCP



RTU



 **BACnet**™

MSTP



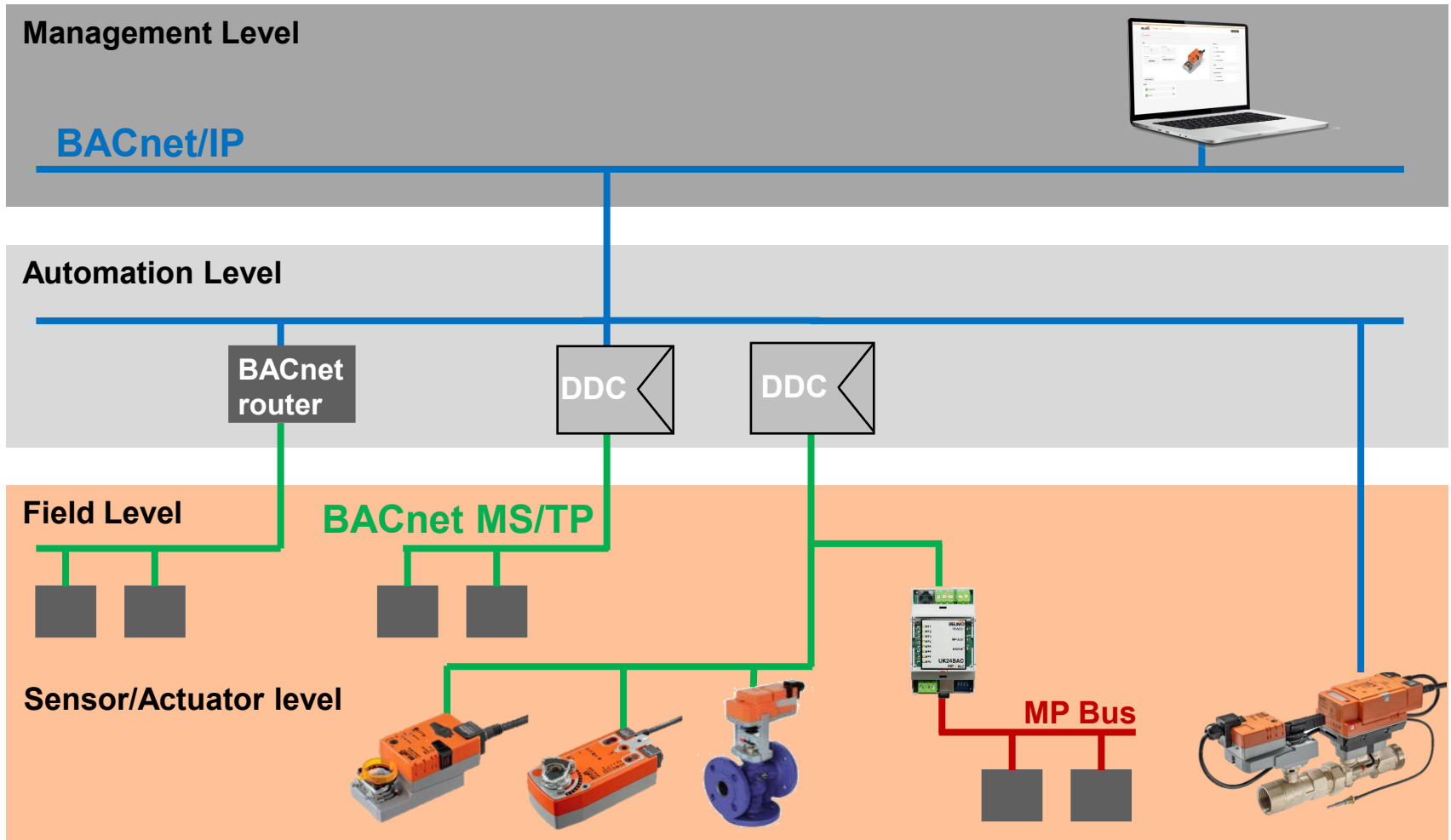
IP



 **KNX**®



BACnet IP and BACnet MS/TP



BACnet IP Addressing



IP Subnet: 192.168.0.1...254
Subnet Mask: 255.255.255.0

InstNr: 10

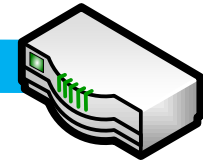
IP Addr: 192.168.0.10



BACnet Client - Yabe

InstNr: 1000

IP Addr: 192.168.0.100



BACnet Router

UDP Port: 0xBAC0 = 47'808

InstNr: 2

VLM24A-LP1

IP Addr: 192.168.0.2



InstNr: 3

EV015R2+BAC

IP Addr: 192.168.0.3



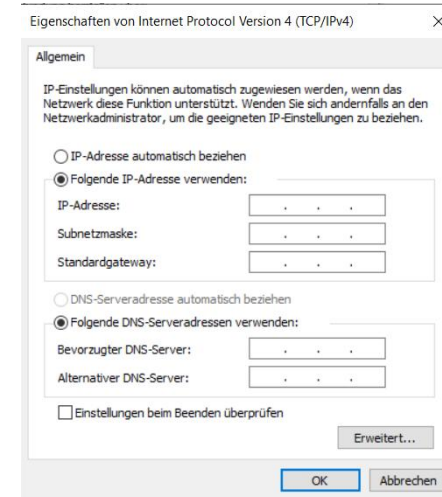
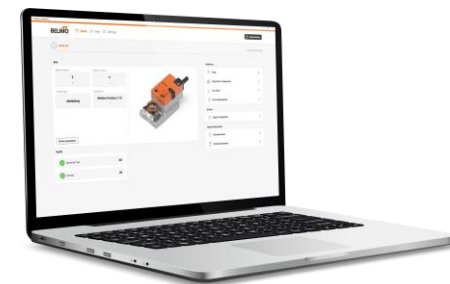
Parameterisation



IP Configuration for Belimo Devices

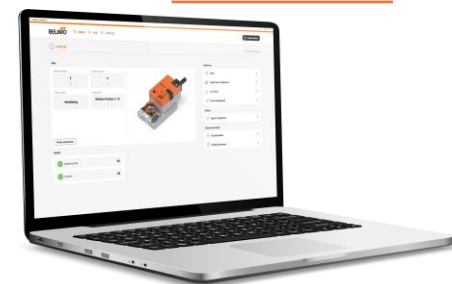
IP Address	For example: 192.168.1.10	IP Address , corresponding to the IP Subnet (IP V4 address)
Subnet Mask	For example: 255.255.255.0	Subnet Mask of the IP Ethernet network (Definition of Subnet Size)
Default Gateway	For example: 192.168.1.50	Default Gateway of the IP Ethernet subnet (Address to which all packets with IP addresses from other subnets should be sent: IP Router to other networks)

➔ *IP Addressing must be configured according to the specific IT Network*



BACnet IP Configuration for Belimo Devices

Device Name	Site4'Floor2 'Zone3'VAV1	BACnet Internetwork-wide unique Text for the Device Identification
Instance Number	1 - 4194302	BACnet Internetwork-wide unique Number for the Device Identification
Port	Default: 47'808	Can be selected free. Default is 47'808 (0xBAC0).



BACnet IP Settings

Port

☒ Simple Device
☐ Foreign Device

Device Object Settings

Instance ID

Device Name

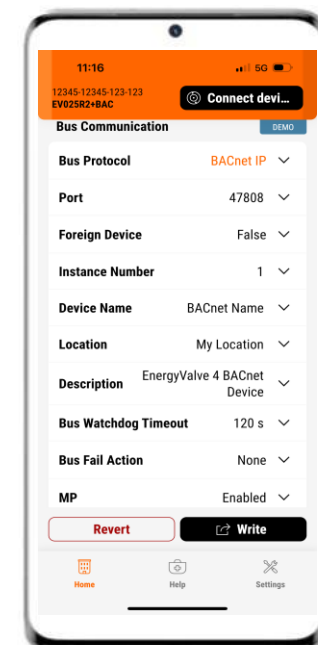
Device Description

Location

System Status

Protocol Version

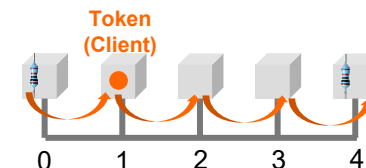
Protocol Revision



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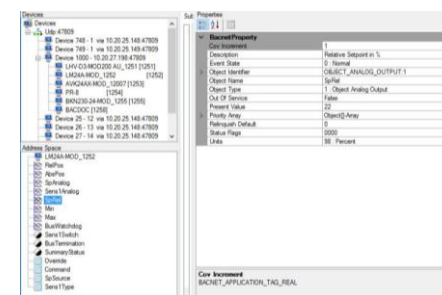
Bus Line:



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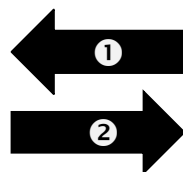
BACnet Objects and Services

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Rel. Position % (AI2)
Setpoint Source (MV220)

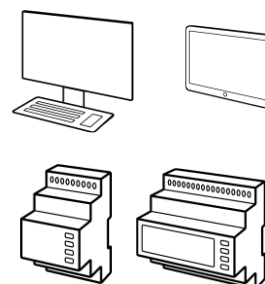


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Read Data
Write Data
Value Change
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Object Type
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Read Property:
Object Name

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- Sens1Type

RelPos (Analog Input Object)

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Description	Relative Position in %
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Status Flags	0000
Units	98 : Percent

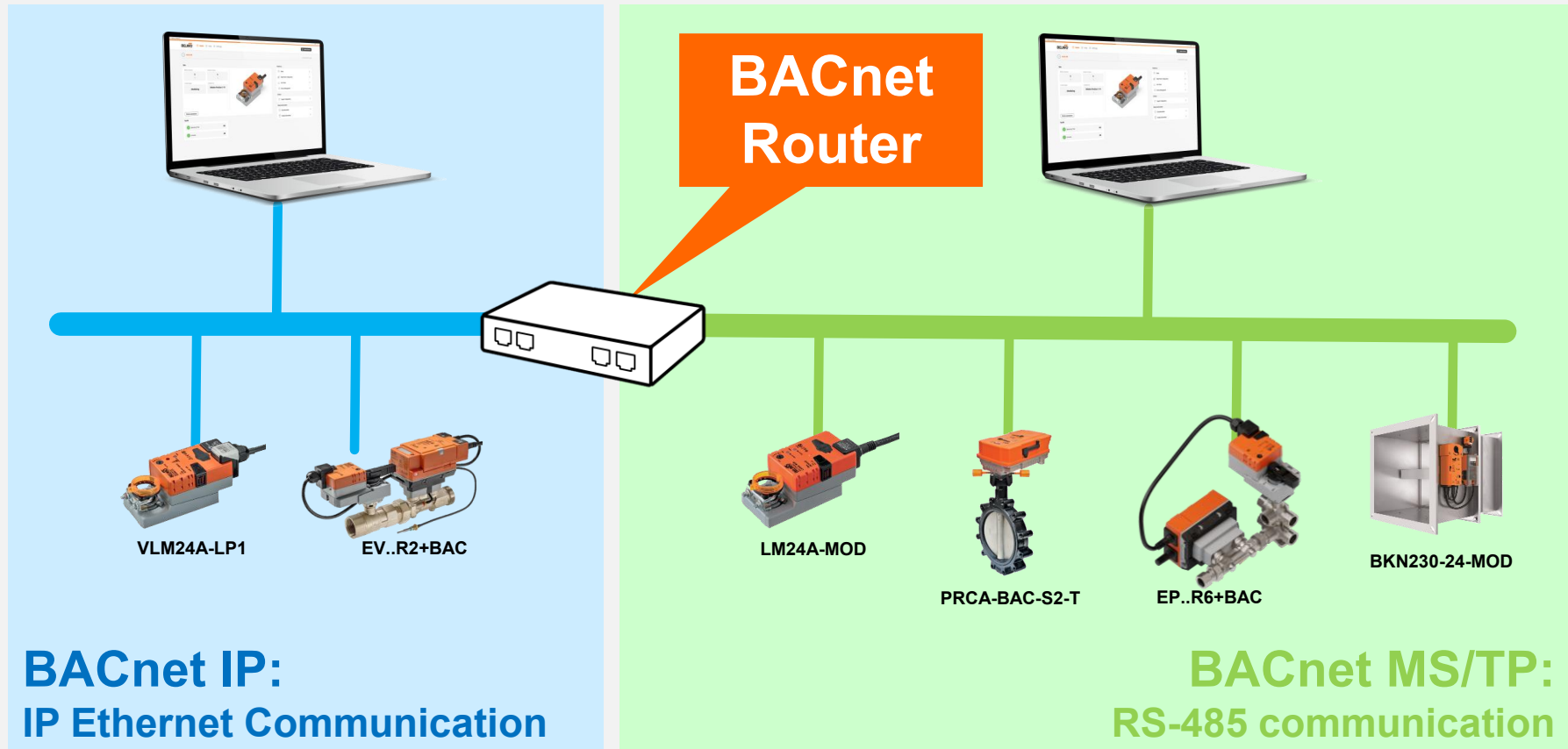
BACnet IP and BACnet MS/TP



BACnet Internetwork

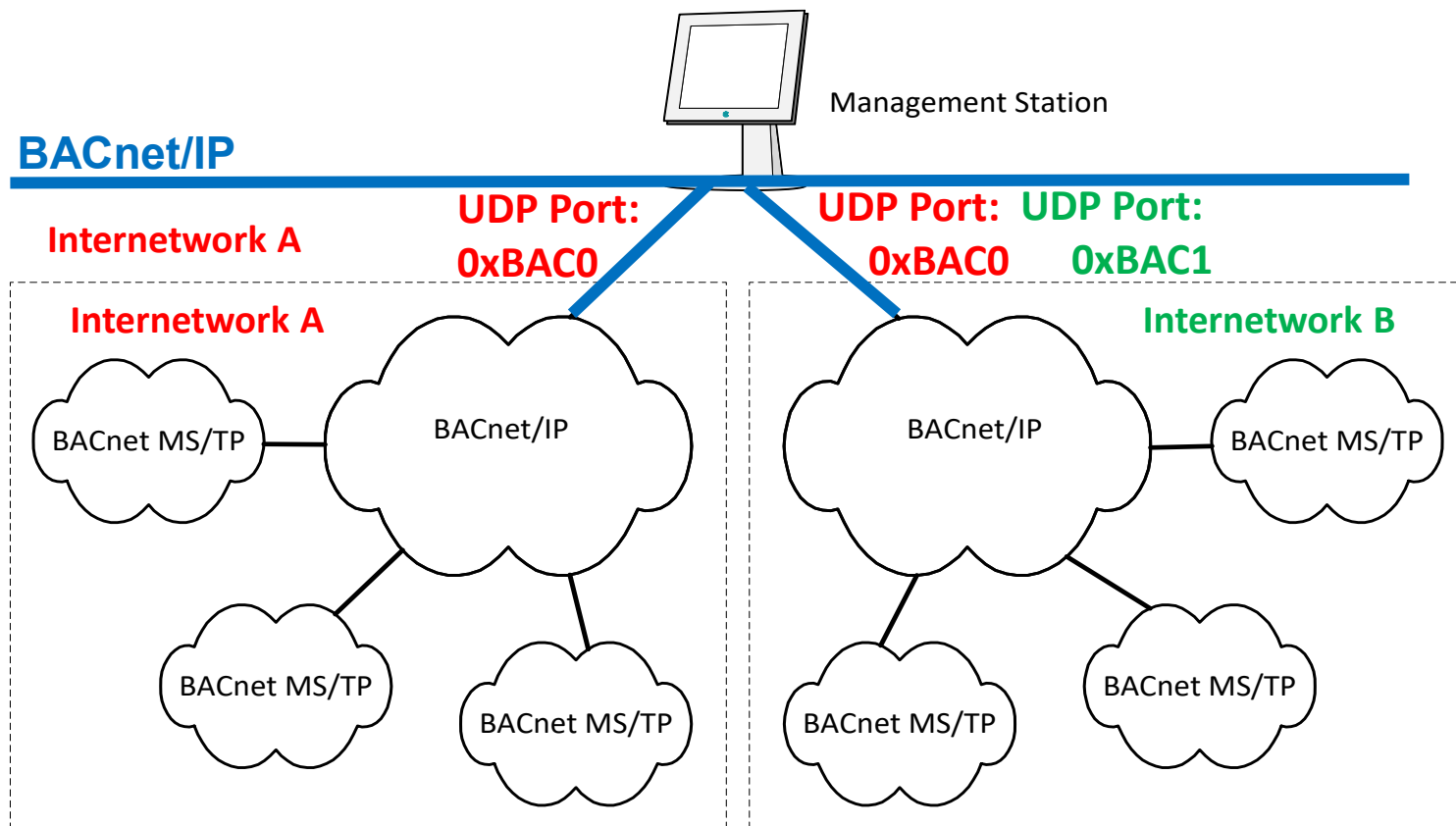


BACnet Instance Number & Device Name must be unique in the entire BACnet Internet network!



BACnet IP Network Design with UDP Ports

Several BACnet Internetworks



Workshop III



Workshop III



- Placeholder -> PPT

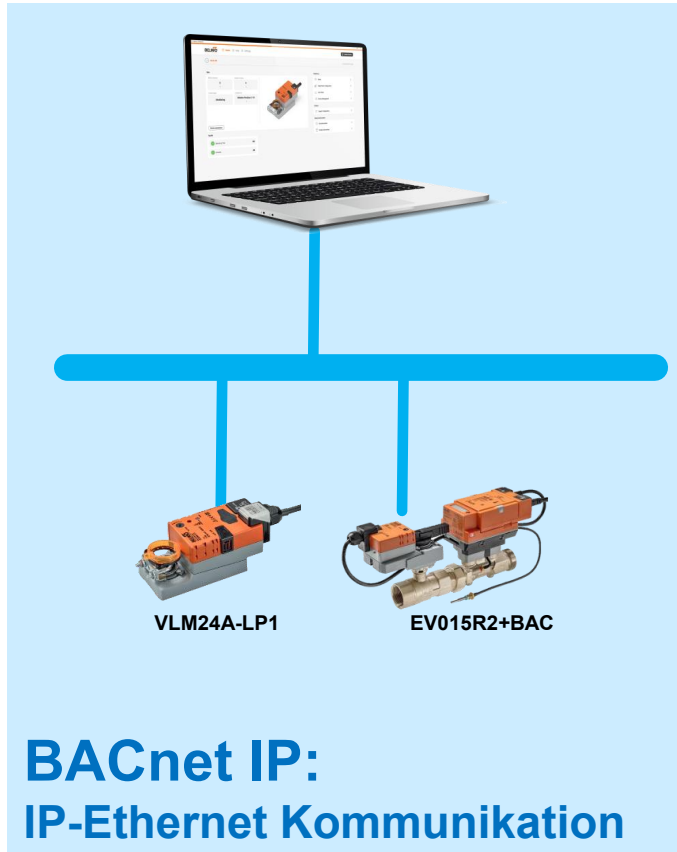
Reset tool settings Yabe after workshop



- Options - Settings:

DefaultUdpPort	47808
DefaultWritePriority	LOWEST_AND_DEFAULT
YabeDeviceId	-1
Udp_DontFragment	False
Udp_ExclusiveUseOfSocket	True

BACnet IP – special therms



BACnet IP:

BACnet IP uses IP Broadcast Messages!

BBMD:

BACnet Broadcast Management Device

FD:

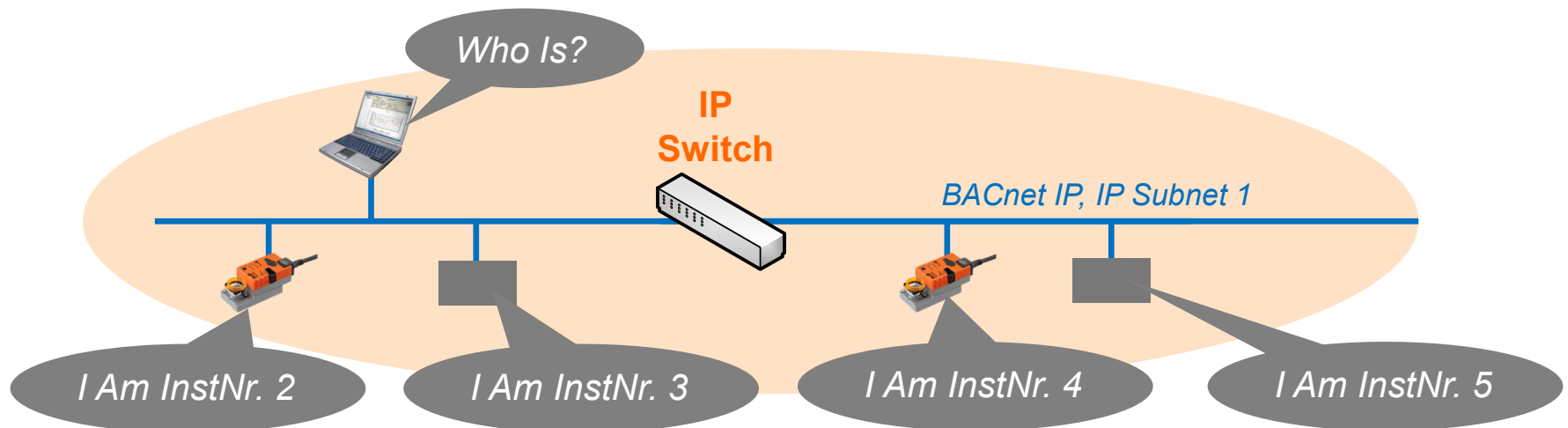
BACnet Foreign Device

BACnet IP: Broadcast Messages with BACnet

Example: Who Is?

1 IP Subnet, BACnet Client starts in the BACnet IP Network:

1. When the **BACnet client starts**, a «**Who Is**» broadcast is first sent to the network to find BACnet devices
2. **All BACnet devices** in this IP subnet respond **directly** to the BACnet client with the message «**I Am InstNr. XX**».

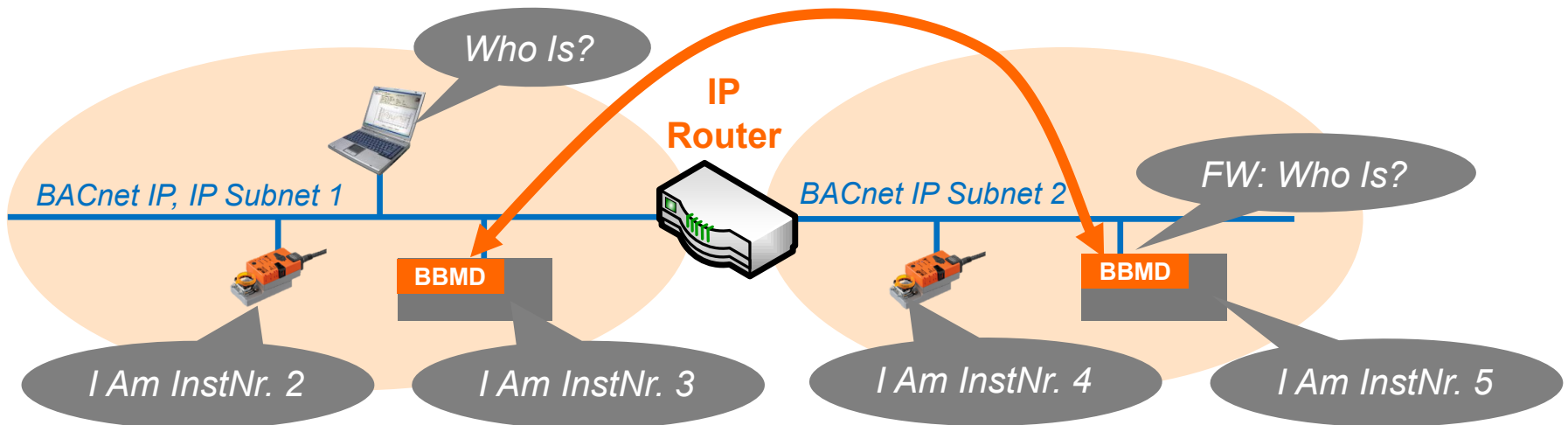


BACnet IP: Broadcast Messages with BACnet

BBMD: BACnet Broadcast Management Device

2 IP Subnet with IP Router, BACnet Client starts in the BACnet IP Network:

1. When the **BACnet client starts**, it first sends a " «Who Is» broadcast to the network to find BACnet devices.
2. The **IP router stops** the broadcast. Only the BACnet devices on the same subnet as the client respond **directly** to the BACnet client with the message «I Am InstNr. XX»
3. Enable a BBMD in both IP subnets: One BBMD sends broadcast to the other BBMD, which distributes the broadcast in its local IP subnet
4. **All BACnet devices in both IP subnets** respond **directly** to the BACnet client with the message «I Am InstNr. XX»

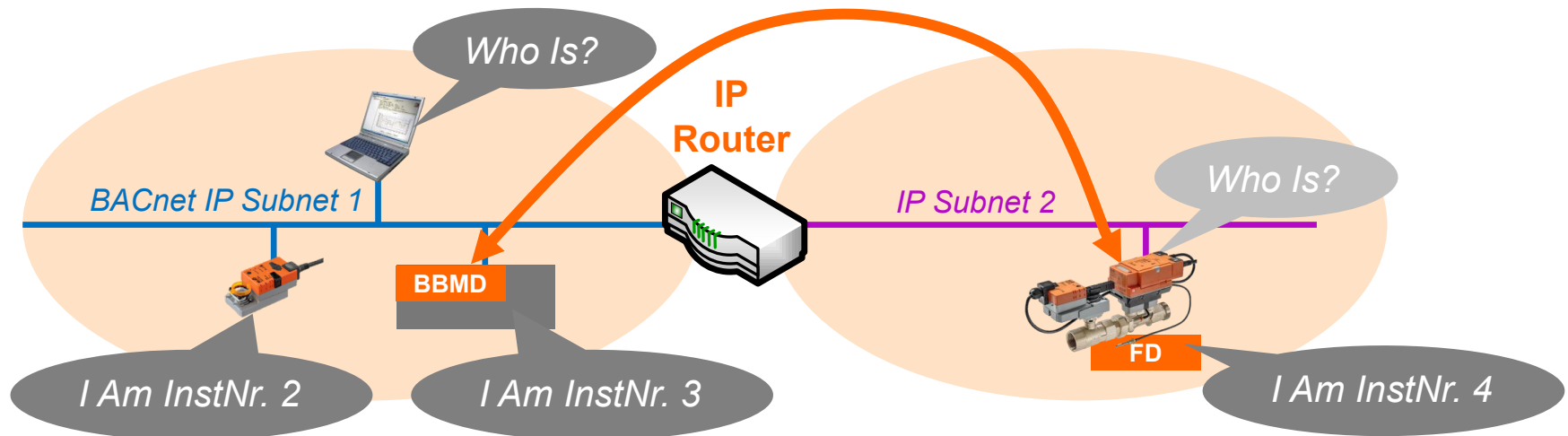


BACnet IP: Broadcast Messages with BACnet

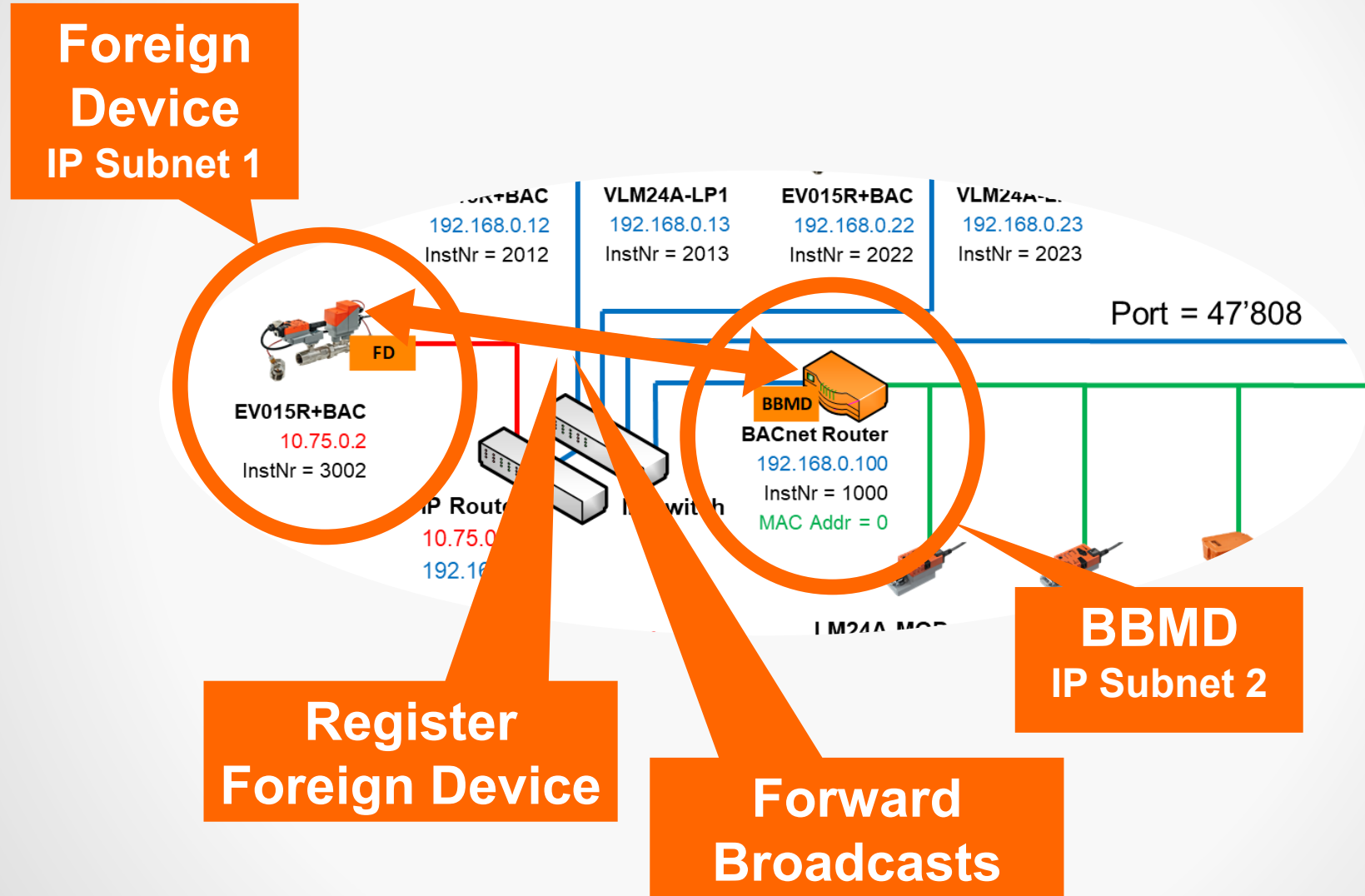
FD: BACnet Foreign Device

2 IP Subnets with IP Router, BACnet Client starts in the BACnet IP Network:

1. When the **BACnet client starts**, it first sends a " «**Who Is**» broadcast to the network to find BACnet devices.
2. The **IP router stops** the broadcast. Only the BACnet devices on the same subnet as the client respond **directly** to the BACnet client with the message «**I Am InstNr. XX**»
3. Enable a BBMD in IP subnet 1, enable FD at the device which is «lonely» and apart from the others. The Foreign Device registers itself at the BBMD to receive Broadcasts.
4. The BBMD forwards Broadcast to the ForeignDevice.
5. **The Foreign Device** responds **directly** to the BACnet client with the message «**I Am InstNr. XX**»



Foreign Device Konfiguration



Configuration for Foreign Device (FD):

- Register BBMD in a BACnet device in BACnet network
 - In our exercise: BBMD is the Router
- Energy Valve in another IP Subnet: Register as Foreign Device

BACnet IP Einstellungen

47808	Port
☐ Einfaches Gerät	
☒ Fremdgerät	
192.168.0.100	IP BBMD
30	Time-to-Live



Übersicht

Daten

Zustand

Einstellungen

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Datum & Zeit

Benutzer

IP

BACnet/MP/Modbus

Cloud

Wartung

BACnet, MP-Bus und Modbus Einstellungen

Kommunikations-Protokoll

- ☒ BACnet IP
- ☐ BACnet MS/TP
- ☐ MP
- ☐ Modbus TCP
- ☐ Modbus RTU
- ☐ Keine

BACnet IP Einstellungen

Port
☐ Einfaches Gerät
☒ Fremdgerät
 IP BBMD
 Time-to-Live

Einstellungen Geräteobjekt

Instance ID
 Gerätename
 Gerätebeschreibung
 Ort
 Systemstatus
 Protocol Version
 Protocol Revision

Ausführen

Questions?





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